

ANKARA GÜVERCİNLİK, Turkey

WMO: 171310

Lat: 39.935N

Lon: 32.741E

Elev: 2694

StdP: 13.32

Time Zone: 3.00 (E03)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.6	17.5	7.6	9.0	19.5	11.5	10.9	23.5	19.9	45.9	17.5	43.9	2.4	220	0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.0	95.0	63.2	91.7	62.8	89.3	62.3	66.5	85.7	65.3	84.5	64.2	83.1	8.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.7	87.7	72.8	59.0	82.3	71.0	57.3	77.5	70.8	32.6	85.5	31.6	84.4	30.7	83.3	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	15.5	13.4	8.3	98.7	5.1	2.2	4.7	100.2	1.7	101.5	-1.2	102.7	-4.9	104.3
			5.5	68.6	6.7	2.3	0.7	70.3	-3.2	71.6	-7.0	72.9	-11.9	74.6
				DB										
				WB										

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	54.4	33.5	37.1	44.4	52.2	61.2	68.8	75.1	75.3	66.5	56.4	44.9	36.8	(6)
(7)		DBStd	15.88	8.54	8.08	7.13	6.55	5.60	5.16	4.38	4.70	5.85	6.68	6.75	7.50	(7)
(8)		HDD50	1742	514	365	198	50	1	0	0	0	24	179	411		(8)
(9)		HDD65	4753	978	782	639	387	145	21	0	2	50	273	602	874	(9)
(10)		CDD50	3360	1	3	24	115	347	564	778	784	494	221	27	2	(10)
(11)		CDD65	899	0	0	0	2	27	135	313	321	95	6	0	0	(11)
(12)		CDH74	11353	0	0	3	58	464	1643	3849	3825	1352	159	0	0	(12)
(13)		CDH80	4745	0	0	0	6	106	574	1783	1786	471	19	0	0	(13)
(14)	Wind	WSAvg	5.1	4.5	4.9	5.9	5.6	5.4	5.8	6.2	5.8	4.8	4.1	3.5	4.1	(14)
(15)	Precipitation	PrecAvg	15.4	1.5	1.2	1.7	1.9	1.9	1.4	0.6	0.7	0.7	1.1	1.2	1.7	(15)
(16)		PrecMax	20.6	3.7	3.1	4.6	3.6	6.8	4.3	2.6	2.6	2.7	2.8	2.5	5.4	(16)
(17)		PrecMin	10.6	0.2	0.4	0.5	0.6	0.3	0.0	0.0	0.0	0.1	0.2	0.1	0.1	(17)
(18)		PrecStd	2.9	0.8	0.7	1.1	1.1	1.4	1.1	0.6	0.7	0.8	0.6	0.7	1.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.7	63.4	71.8	80.5	87.4	93.2	99.4	98.7	93.4	82.3	70.0	59.1	(19)
(20)		0.4%	MCWB	45.5	47.9	50.2	55.8	59.2	62.5	63.5	63.6	60.9	57.7	53.0	49.2	(20)
(21)		2%	DB	51.6	57.5	66.2	74.5	82.7	89.4	95.2	95.1	89.1	78.5	66.2	53.9	(21)
(22)		2%	MCWB	43.9	46.4	49.0	54.0	58.7	62.1	63.5	63.4	60.1	56.7	52.1	46.4	(22)
(23)		5%	DB	48.1	53.9	62.4	70.2	79.1	86.1	91.7	91.8	85.8	75.0	62.3	51.0	(23)
(24)		5%	MCWB	42.7	44.2	47.4	52.4	58.0	61.5	63.0	63.0	59.2	55.9	50.1	45.1	(24)
(25)		10%	DB	44.9	50.1	58.7	66.5	75.4	82.8	89.2	89.4	82.1	71.0	57.6	48.1	(25)
(26)		10%	MCWB	40.9	43.0	46.2	50.9	56.7	60.5	62.6	62.7	58.5	54.7	48.2	43.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.2	50.6	52.1	57.7	62.5	66.0	68.4	69.0	64.4	60.1	56.2	49.9	(27)
(28)		0.4%	MCDWB	53.7	60.6	66.5	73.1	78.9	84.5	88.0	87.2	83.3	73.1	65.8	57.1	(28)
(29)		2%	WB	44.8	47.2	50.1	55.6	60.7	64.4	66.5	66.8	62.5	58.7	53.6	47.5	(29)
(30)		2%	MCDWB	50.5	55.8	62.0	70.4	76.5	82.2	86.1	85.5	80.5	71.5	62.7	52.7	(30)
(31)		5%	WB	42.8	45.0	48.4	54.0	59.3	63.2	65.3	65.6	61.2	57.5	51.5	45.8	(31)
(32)		5%	MCDWB	46.7	52.5	59.4	67.0	74.8	80.5	84.9	84.7	78.7	70.4	59.7	50.2	(32)
(33)		10%	WB	41.0	43.1	47.0	52.3	58.1	62.0	64.0	64.5	60.1	56.1	49.0	43.9	(33)
(34)		10%	MCDWB	44.6	49.1	56.9	64.3	72.6	78.6	83.5	83.4	77.0	68.1	56.6	47.7	(34)
(35)	Mean Daily Temperature Range		MDBR	13.7	17.8	20.5	22.1	23.4	24.6	26.0	25.7	26.7	23.9	21.8	14.7	(35)
(36)		5% DB	MCDBR	19.1	24.7	28.4	28.8	29.4	29.3	30.1	29.4	32.4	30.7	28.3	19.0	(36)
(37)			MCWBR	12.4	14.4	14.0	12.6	10.8	9.4	9.1	9.0	11.7	13.6	15.3	12.3	(37)
(38)			MCDBR	15.8	21.2	23.6	24.9	25.2	26.0	26.7	25.5	26.5	24.9	23.6	16.9	(38)
(39)		5% WB	MCWBR	10.8	12.8	12.3	11.8	10.1	9.1	8.9	8.6	10.2	11.6	13.2	11.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.307	0.333	0.376	0.412	0.392	0.372	0.369	0.367	0.348	0.348	0.317	0.307	(40)	
(41)		taud	2.381	2.327	2.225	2.150	2.275	2.351	2.355	2.338	2.392	2.389	2.447	2.438	(41)	
(42)		Ebn at Noon	271	279	277	273	281	286	286	284	283	270	266	262	(42)	
(43)		Edh at Noon	27	33	41	47	42	39	39	38	34	30	25	24	(43)	
(44)	All-Sky Solar Radiation	RadAvg	595	914	1279	1640	1951	2219	2351	2094	1650	1127	779	529	(44)	
(45)		RadStd	73	132	114	146	139	184	98	84	111	81	77	68	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page